

MF20006 Introduction to Computer Science

Course Introduction

Hui Xu

xuh@fudan.edu.cn



Instruction Team

□ Instructor: XU, Hui

- Ph. D. degree from CUHK
- Research Interests: program analysis, software reliability
- Email: xuh@fudan.edu.cn
- Office: Room D6023, X2 Building, Jiangwan Campus



□ Tutor: Yilin Chen

- Email: yilincheng25@m.fudan.edu.cn



Objective of This Course

- ❑ **Master AI tools in learning and solving problems.**
- ❑ Understand how computers work.
- ❑ Think computationally.
- ❑ Be able to read code.
- ❑ Develop tools using coding agent.

Tentative Schedule

Week	Date	Phase	Course
1	Sep-10	Hardware and Networks	Numbers and Computation
2	Sep-17		Computer Architecture
3	Sep-24		The Internet and the Web
4	Oct-1		Holiday
5	Oct-8	Programming and Algorithms	Programming Languages and Compilers
6	Oct-15		Data Structures
7	Oct-22		Algorithms - I
8	Oct-29		Algorithms - II
9	Nov-5	AI and Application Software	Artificial Intelligence - I
10	Nov-12		Artificial Intelligence - II
11	Nov-19		Application Software - Frontend
12	Nov-26		Application Software - Backend
13	Dec-3	Miscellaneous Topics	Project Proposal
14	Dec-10		Multimedia and Signals
15	Dec-17		Cryptography
16	Dec-24		Project Report



Reference Book

- ❑ Current textbooks cannot keep pace with the latest advances in AI.
- ❑ We encourage students to explore and learn by prompting AI.

No Textbook Required

Course Information

❑ Time: Thursday 6:30pm - 9:05pm

❑ Classroom: H6305

❑ Platform:

➤ Course Webpage: eLearning

- Discussion
- Lecture notes
- Project report submission

➤ External Webpage: <https://intro2cs.github.io/>

➤ Notification: WeChat Group

Grading

❑ In-Class Participation: 20%

- Participate in classroom discussions and activities (or via eLearning during class).
- Ask at least five meaningful questions or provide meaningful responses to questions raised by others.

❑ Project: 30%

- 1-3 students form a group
- Project proposal: 13th week (5%)
- Project report: 16th week (25%)

❑ Final Exam: 50%

- Closed-book
- Time: 2026-12-31 15:30~17:30
- Classroom: TBA

Discussion Platform: eLearning

2026-2027 学年第一学期

Home

Discussions

Grades

People

Syllabus

L1 Numbers and Computation
徐辉
All Sections

Search entries or author

Unread

✓ Subscribe

Edit View Insert Format Tools Table

12pt Paragraph B I U A T²

Post your questions here.

p

Attach

0 words

Cancel Post Reply

Project

❑ Scope:

- Develop a tool or system that solves a real-world problem.
- Conduct data analytics or a survey related to computer science.

❑ A general structure for reference:

➤ Problem:

- What problem are you trying to solve?
- Why is this problem worth investigation?

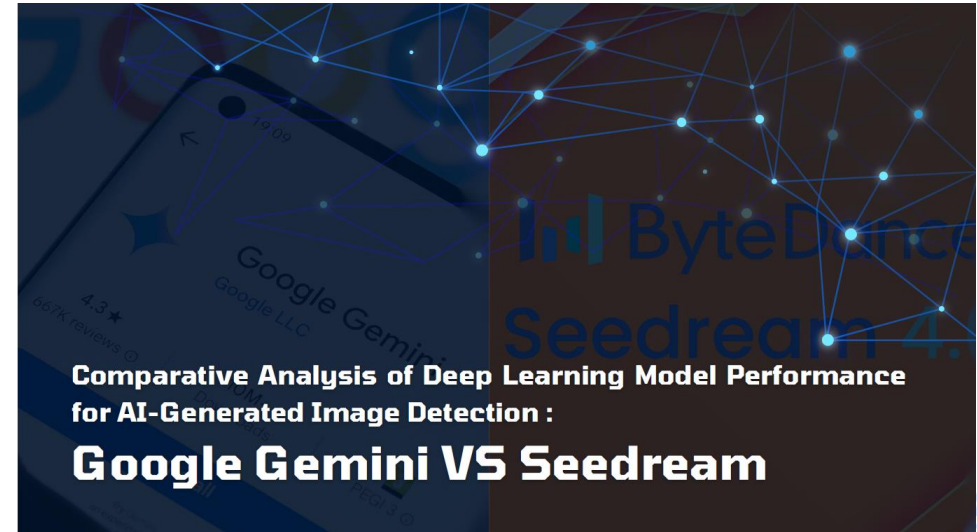
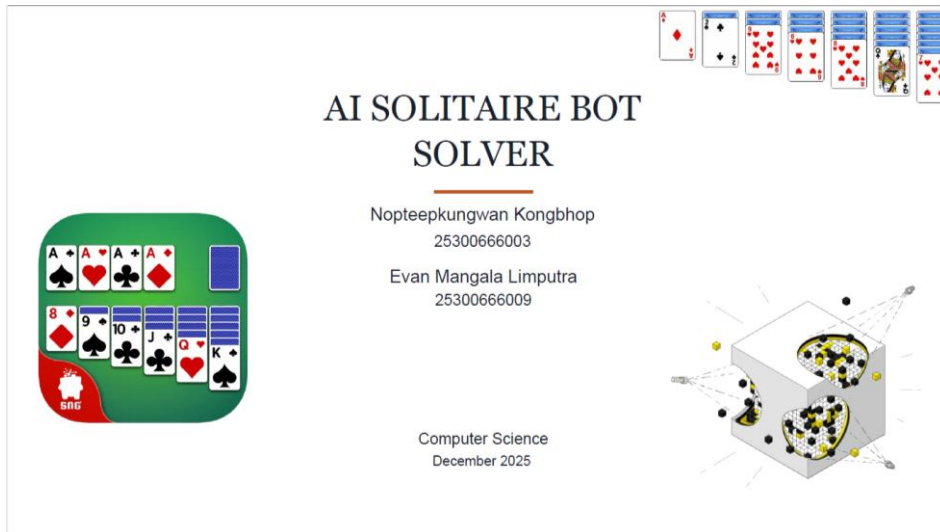
➤ Approach:

- What is your proposed solution?
- What is the detailed design of your solution?

➤ Evaluation:

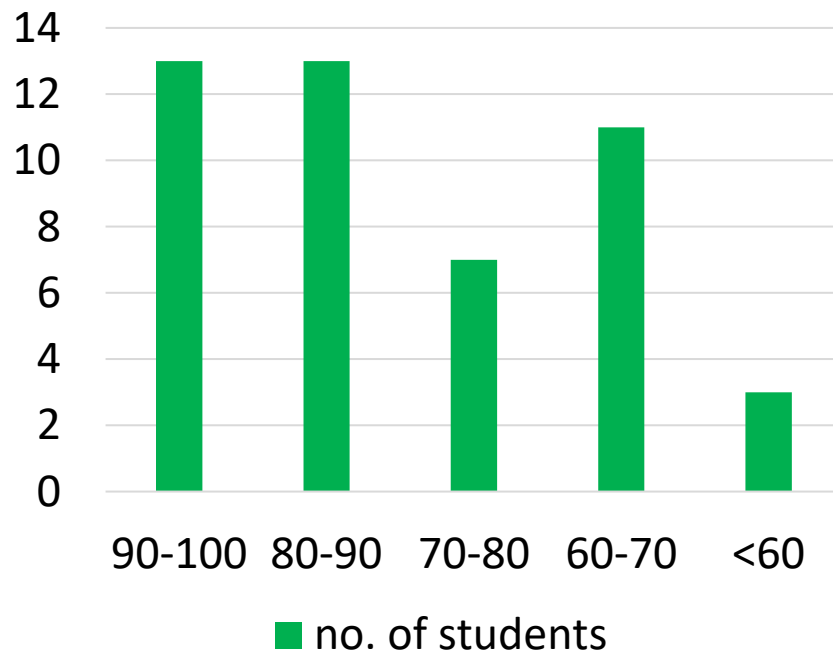
- For a new tool: Demonstrate that your approach can effectively address the target problem.
- For a survey: Demonstrate how your study can answer the research questions.

Sample Projects

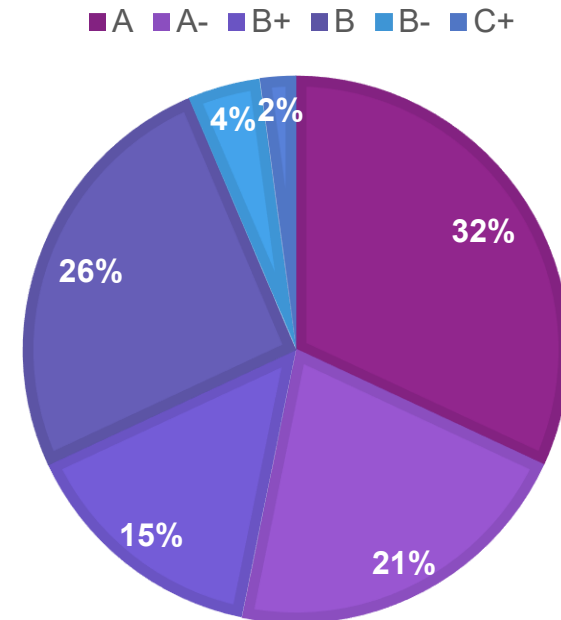


Distribution of Grade

Final Exam



Grade



Plagiarism Will Not Be Tolerated

❑ **Students are encouraged to use AI tools as an assistant.**

- Search for useful materials
- Agentic coding

❑ **However, you are responsible for everything you submit.**

- Do not submit anything you do not understand or cannot explain.
- Any use of others' work must be properly quoted or cited.

